# Graphs of Sine and Cosine Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.03

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Amplitude, period, and midline**

For the function $y = 3\sin(2x)$, state:

**a** the amplitude $= 3$ **b** the period $= \pi$ **c** the range $[-3, 3]$

(a) $= 3$; (b) $= \pi$; (c) $[-3, 3]$

**2** **[Foundational]** • **Target: Amplitude, period, and midline**

For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:

**a** the amplitude $= 2$

**b** the period $= 4\pi$

**c** the midline $y = 1$

**d** the range $[-1, 3]$

(a) $= 2$; (b) $= 4\pi$; (c) $y = 1$; (d) $[-1, 3]$

**3** **[Foundational]** • **Target: Amplitude, period, and midline**

For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.

Phase shift: $\frac{\pi}{3}$ to the right. Period: $2\pi$.

**4** **[Foundational]** • **Target: Reading and writing equations from graphs**

The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.

Amplitude 2, period $2\pi$. Equation: $y = 2\sin x$.

**5** **[Foundational]** • **Target: Reading and writing equations from graphs**

Write an equation of a cosine function with amplitude 4, period $\pi$, reflected across the $x$-axis, with no phase or vertical shift.

$y = -4\cos(2x)$ (period $\frac{2\pi}{b} = \pi$ gives $b = 2$).

© ClipSAT — clipsat.org | All Rights Reserved

**6** [Foundational] • Target: Reading and writing equations from graphs

The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where $t$ is in hours after midnight.

**a** State the maximum and minimum depths. Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.

**b** State the period of the tide cycle. Period $= \frac{2\pi}{\pi/6} = 12$ hours.

**c** Find the depth at $t = 3$. $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.

(a) Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.; (b) Period $= \frac{2\pi}{\pi/6} = 12$ hours.; (c) $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.

**7** [Foundational] • Target: Graphing with transformations

The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.

Midline $y = 1$; maximum value $2$; minimum value $0$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Graphing with transformations

Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.

Horizontal compression by a factor of $2$ (period $\pi$), reflection across the $x$-axis, vertical stretch by a factor of $3$, shift right $\frac{\pi}{4}$, and shift up $2$.

**9** [Intermediate] • Target: Graphing with transformations

State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

Amplitude $5$; period $\frac{2\pi}{3}$; phase shift $\frac{\pi}{6}$ to the left; midline $y = -4$.

© ClipSAT — clipsat.org | All Rights Reserved

**10** [Intermediate] • Target: Modeling periodic phenomena

The average monthly temperature in a city varies between $8°\text{C}$ (in January, month $1$) and $28°\text{C}$ (in July, month $7$), following a sinusoidal pattern with a 12-month period.

**a** Find the amplitude and midline of the model. Amplitude $= \frac{28-8}{2} = 10$; midline $= \frac{28+8}{2} = 18$.

**b** Write a cosine model $T(m)$ for the temperature in month $m$, using the fact that the minimum occurs at $m = 1$. Since the minimum is at $m = 1$, use a negative cosine shifted:
$T(m) = -10\cos\left(\frac{2\pi}{12}(m-1)\right) + 18 = -10\cos\left(\frac{\pi}{6}(m-1)\right) + 18$.

**c** Use your model to estimate the temperature in month $4$ (April).
$T(4) = -10\cos\left(\frac{\pi}{6}(3)\right) + 18 = -10\cos\left(\frac{\pi}{2}\right) + 18 = 18°\text{C}$.

(a) Amplitude $= \frac{28-8}{2} = 10$; midline $= \frac{28+8}{2} = 18$.; (b) Since the minimum is at $m = 1$, use a negative cosine shifted: $T(m) = -10\cos\left(\frac{2\pi}{12}(m-1)\right) + 18 = -10\cos\left(\frac{\pi}{6}(m-1)\right) + 18$.; (c) $T(4) = -10\cos\left(\frac{\pi}{6}(3)\right) + 18 = -10\cos\left(\frac{\pi}{2}\right) + 18 = 18°\text{C}$.

**11** [Intermediate] • Target: Comparing sine and cosine graphs

Explain why $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ by comparing the graphs of sine and cosine.

The cosine graph looks identical to the sine graph shifted $\frac{\pi}{2}$ units to the LEFT. Shifting $\sin x$ left by $\frac{\pi}{2}$ means replacing $x$ with $x + \frac{\pi}{2}$, giving $\sin\left(x + \frac{\pi}{2}\right) = \cos x$.

**12** [Intermediate] • Target: Comparing sine and cosine graphs

Both $y = 2\sin(x)$ and $y = 2\cos(x)$ are graphed below. State one key visual difference and one similarity between them.

Similarity: both have the same amplitude $(2)$ and period $(2\pi)$. Difference: the sine curve passes through the origin $(0, 0)$, while the cosine curve reaches its maximum at $x = 0$ — they are horizontally shifted versions of the same shape.

**13** [Intermediate] • Target: Finding zeros and extrema from an equation

Find all zeros of $y = \sin(2x)$ on the interval $[0, 2\pi]$.

$2x = 0, \pi, 2\pi, 3\pi, 4\pi \Rightarrow x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.

**14** [Intermediate] • Target: Finding zeros and extrema from an equation

Find the $x$-value(s) in $[0, 2\pi]$ where $y = 3\cos(x)$ attains its maximum and minimum values.

Maximum value $3$ occurs at $x = 0$ and $x = 2\pi$ (where $\cos x = 1$). Minimum value $-3$ occurs at $x = \pi$ (where $\cos x = -1$).

© ClipSAT — clipsat.org | All Rights Reserved

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 7.02 Logarithms and Their Properties

Solve $2\sin(x) + 1 = 0$ for $x \in [0, 2\pi]$.

$\sin x = -\frac{1}{2}$: $x = \frac{7\pi}{6}$ or $x = \frac{11\pi}{6}$.

**16** [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A Ferris wheel's height above ground is $h(t) = 15\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) + 17$ meters. Find the first time $t > 0$ (minutes) when the rider is at height $17$ m and rising.

Height $= 17$ (the midline) when $\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) = 0$. The rider is rising through the midline when the argument equals $0$: $\frac{\pi}{10}t - \frac{\pi}{2} = 0 \Rightarrow t = 5$ minutes.

**17** [Challenging] • Target: Finding zeros and extrema from an equation • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the period and phase shift of $y = \tan(3x - 1)$.

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = \frac{1}{3}\pi$, phase shift $= \frac{C}{B} = \frac{1}{3}$ units right. Final Answer: Period $\frac{1}{3}\pi$, phase shift $\frac{1}{3}$ units right

**18** [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of $a$.

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

**19** [Challenging] • Target: Amplitude and Vertical Shift • Cross-Lesson: 8.02 Graphing Rational Functions

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by $r$: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on $x$: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x - 2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

© ClipSAT — clipsat.org | All Rights Reserved

**20** [Challenging] • Target: Period and Frequency • Cross-Lesson: 8.03 Introduction to Limits

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$.
$(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Amplitude and Vertical Shift

Find the amplitude and midline equation of $y = -4\cos(x) + 3$.

**a.** **Amplitude $= 4$, Midline: $y = 3$** ✓

**b.** Amplitude $= -4$, Midline: $y = 3$

**c.** Amplitude $= 4$, Midline: $y = -3$

**d.** Amplitude $= 8$, Midline: $y = 3$

For $y = A\cos(x) + D$, amplitude is $|A| = |-4| = 4$, and the midline is the horizontal line $y = D = 3$.

**22** [Foundational] • Target: Amplitude and Vertical Shift

What are the maximum and minimum values of $f(x) = 5\sin(x) - 2$?

**a.** **Maximum $= 3$, Minimum $= -7$** ✓

**b.** Maximum $= 5$, Minimum $= -5$

**c.** Maximum $= 7$, Minimum $= -3$

**d.** Maximum $= 3$, Minimum $= -3$

$\sin(x) \in [-1, 1]$. Maximum is $5(1) - 2 = 3$. Minimum is $5(-1) - 2 = -7$.

**23** [Foundational] • Target: Amplitude and Vertical Shift

A wave oscillates between a maximum of 12 and a minimum of 4. Find its amplitude.

**a.** 4 ✓

**b.** 8

**c.** 6

**d.** 16

Amplitude $= \frac{\text{Max} - \text{Min}}{2} = \frac{12 - 4}{2} = \frac{8}{2} = 4$.

© ClipSAT — clipsat.org | All Rights Reserved

**24** [Foundational] • Target: Period and Frequency

Find the period of the function $y = 3\sin(4x)$.

**a.** $\frac{\pi}{2}$ ✓

**b.** $\pi$

**c.** $2\pi$

**d.** $\frac{\pi}{4}$

Period $T = \frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$.

**25** [Foundational] • Target: Period and Frequency

Find the frequency (cycles per unit of $x$) of $y = \cos\left(\frac{\pi}{3}x\right)$.

**a.** 1/6 ✓

**b.** 6

**c.** $\pi/3$

**d.** 3

Period $T = \frac{2\pi}{\pi/3} = 6$. Frequency $= \frac{1}{T} = \frac{1}{6}$.

**26** [Foundational] • Target: Period and Frequency

Write a sine function with amplitude 3 and period $4\pi$.

**a.** $y = 3\sin\left(\frac{1}{2}x\right)$ ✓

**b.** $y = 3\sin(4x)$

**c.** $y = 3\sin(2x)$

**d.** $y = 4\sin\left(\frac{1}{3}x\right)$

Period $4\pi = \frac{2\pi}{B} \Rightarrow B = \frac{2\pi}{4\pi} = \frac{1}{2}$. Function: $y = 3\sin\left(\frac{1}{2}x\right)$.

**27** [Foundational] • Target: Phase Shift (Horizontal Shift)

Find the phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

**a.** $\frac{\pi}{6}$ **to the right** ✓

**b.** $\frac{\pi}{2}$ to the right

**c.** $\frac{\pi}{6}$ to the left

**d.** $\frac{3\pi}{2}$ to the right

Rewrite as $2\sin\left(3\left(x - \frac{\pi}{6}\right)\right)$. Phase shift $C/B = \frac{\pi/2}{3} = \frac{\pi}{6}$ to the right.

© ClipSAT — clipsat.org | All Rights Reserved

**28** [Foundational] • Target: Phase Shift (Horizontal Shift)

At what value of $x$ does the first cycle start for $y = \cos(2x + \pi)$?

**a.** $-\frac{\pi}{2}$ ✓

**b.** $\pi$

**c.** $-\pi$

**d.** $\frac{\pi}{2}$

Set argument to 0: $2x + \pi = 0 \Rightarrow 2x = -\pi \Rightarrow x = -\pi/2$.

---

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Phase Shift (Horizontal Shift)

Which transformation shifts the graph of $y = \cos x$ to become identical to $y = \sin x$?

**a.** **Shift right by $\frac{\pi}{2}$** ✓

**b.** Shift left by $\frac{\pi}{2}$

**c.** Reflect across the $x$-axis

**d.** Shift up by 1

Since $\cos(x - \pi/2) = \sin x$, shifting $y = \cos x$ to the right by $\pi/2$ gives $y = \sin x$.

**30** [Intermediate] • Target: Modeling Periodic Phenomena

The temperature in a greenhouse follows $T(t) = 15\sin\left(\frac{\pi}{12}(t - 6)\right) + 65$, where $t$ is hours after midnight. What is the maximum temperature?

**a.** $80\,^\circ\text{F}$ ✓

**b.** $65\,^\circ\text{F}$

**c.** $75\,^\circ\text{F}$

**d.** $95\,^\circ\text{F}$

Max temperature $= \text{Midline} + \text{Amplitude} = 65 + 15 = 80\,^\circ\text{F}$.

**31** [Intermediate] • Target: Modeling Periodic Phenomena

In the greenhouse model above, at what time of day ($t$) does the maximum temperature occur?

**a.** $t = 12$ **(12:00 PM / Noon)** ✓

**b.** $t = 6$ (6:00 AM)

**c.** $t = 18$ (6:00 PM)

**d.** $t = 24$ (Midnight)

Sine attains its maximum of 1 when the argument is $\pi/2$: $\frac{\pi}{12}(t - 6) = \frac{\pi}{2} \Rightarrow t - 6 = 6 \Rightarrow t = 12$ (Noon).

© ClipSAT — clipsat.org | All Rights Reserved

**32** [Intermediate] • Target: Modeling Periodic Phenomena

A Ferris wheel of radius $30$ ft has its lowest point $5$ ft off the ground and completes one rotation every 60 seconds. What is the midline height?

**a.** $35$ ft ✓

**b.** $30$ ft

**c.** $65$ ft

**d.** $40$ ft

Lowest point $= 5$ ft, highest point $= 5 + 2(30) = 65$ ft. Midline $= 5 + 30 = 35$ ft.

**33** [Intermediate] • Target: Reading and writing equations from graphs

Find the period and phase shift of $y = \tan(2x - 1)$.

**a.** **Period $\frac{1}{2}\pi$, phase shift $\frac{1}{2}$ units right** ✓

**b.** Period $\frac{1}{2}\pi$, phase shift $\frac{1}{2}$ units left

**c.** Period $\frac{1}{2}\pi$, phase shift $0$

**d.** Period $\pi$, phase shift $\frac{1}{2}$ units right

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = \frac{1}{2}\pi$, phase shift $= \frac{C}{B} = \frac{1}{2}$ units right.

**34** [Intermediate] • Target: Graphing with transformations

Find the period and phase shift of $y = \tan(1/2x - 1)$.

**a.** Period $2\pi$, phase shift $0$

**b.** Period $2\pi$, phase shift $2$ units left

**c.** Period $\pi$, phase shift $2$ units right

**d.** **Period $2\pi$, phase shift $2$ units right** ✓

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = 2\pi$, phase shift $= \frac{C}{B} = 2$ units right.

© ClipSAT — clipsat.org | All Rights Reserved

**35** [Intermediate] • Target: Modeling periodic phenomena

Find the period and phase shift of $y = \tan(1/2x - 2)$.

**a.** Period $2\pi$, phase shift $0$

**b.** Period $\pi$, phase shift $4$ units right

**c.** **Period $2\pi$, phase shift $4$ units right ✓**

**d.** Period $2\pi$, phase shift $4$ units left

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = 2\pi$, phase shift $= \frac{C}{B} = 4$ units right.

**36** [Intermediate] • Target: Comparing sine and cosine graphs

Find the period and phase shift of $y = \tan(x)$.

**a.** Period $2\pi$, phase shift $0$

**b.** Period $\pi$, phase shift $0$

**c.** Period $\pi$, phase shift $0$ units left

**d.** **Period $\pi$, phase shift $0$ units right ✓**

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = \pi$, phase shift $= \frac{C}{B} = 0$ units right.

**37** [Intermediate] • Target: Finding zeros and extrema from an equation

Find the period and phase shift of $y = \tan(x - 1)$.

**a.** **Period $\pi$, phase shift $1$ units right ✓**

**b.** Period $\pi$, phase shift $1$ units left

**c.** Period $\pi$, phase shift $0$

**d.** Period $2\pi$, phase shift $1$

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = \pi$, phase shift $= \frac{C}{B} = 1$ units right.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

**38** [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 8.01 Rational Functions and Asymptotes

What is the amplitude and period of $y = 5\sin(3x) - 3$?

**a.** Amplitude -3, period $\frac{2}{3}\pi$

**b.** **Amplitude 5, period $\frac{2}{3}\pi$** ✓

**c.** Amplitude 5, period $\frac{3}{2}\pi$

**d.** Amplitude 5, period $2\pi$

Amplitude $= |A| = 5$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

**39** [Challenging] • Target: Amplitude and Vertical Shift • Cross-Lesson: 8.02 Graphing Rational Functions

What is the amplitude and period of $y = 3\sin(3x) + 5$?

**a.** Amplitude 3, period $\frac{3}{2}\pi$

**b.** Amplitude 5, period $\frac{2}{3}\pi$

**c.** Amplitude 3, period $2\pi$

**d.** **Amplitude 3, period $\frac{2}{3}\pi$** ✓

Amplitude $= |A| = 3$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

**40** [Challenging] • Target: Period and Frequency • Cross-Lesson: 8.03 Introduction to Limits

What is the amplitude and period of $y = 1\sin(3x) + 2$?

**a.** Amplitude 1, period $2\pi$

**b.** Amplitude 2, period $\frac{2}{3}\pi$

**c.** **Amplitude 1, period $\frac{2}{3}\pi$** ✓

**d.** Amplitude 1, period $\frac{3}{2}\pi$

Amplitude $= |A| = 1$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

© ClipSAT — clipsat.org | All Rights Reserved

**41** [Challenging] • Target: Phase Shift (Horizontal Shift) • Cross-Lesson: 9.01 Arithmetic Sequences and Series

What is the amplitude and period of $y = 1\sin(3x) - 3$?

**a.** Amplitude 1, period $2\pi$

**b.** Amplitude -3, period $\frac{2}{3}\pi$

**c.** Amplitude 1, period $\frac{3}{2}\pi$

**d.** **Amplitude 1, period $\frac{2}{3}\pi$** ✓

Amplitude $= |A| = 1$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

**42** [Challenging] • Target: Modeling Periodic Phenomena • Cross-Lesson: 9.02 Geometric Sequences and Series

What is the amplitude and period of $y = 1\sin(3x)$?

**a.** Amplitude 1, period $2\pi$

**b.** **Amplitude 1, period $\frac{2}{3}\pi$** ✓

**c.** Amplitude 1, period $\frac{3}{2}\pi$

**d.** Amplitude 0, period $\frac{2}{3}\pi$

Amplitude $= |A| = 1$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

**43** [Challenging] • Target: Amplitude, period, and midline • Cross-Lesson: 9.03 The Binomial Theorem

What is the amplitude and period of $y = 3\sin(3x) - 3$?

**a.** Amplitude 3, period $\frac{3}{2}\pi$

**b.** **Amplitude 3, period $\frac{2}{3}\pi$** ✓

**c.** Amplitude -3, period $\frac{2}{3}\pi$

**d.** Amplitude 3, period $2\pi$

Amplitude $= |A| = 3$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

© ClipSAT — clipsat.org | All Rights Reserved

**44** [Challenging] • Target: Reading and writing equations from graphs • Cross-Lesson: 1.01 Functions and Function Notation

What is the amplitude and period of $y = 2\sin(3x) - 3$?

**a. Amplitude 2, period $\frac{2}{3}\pi$ ✓**

**b.** Amplitude -3, period $\frac{2}{3}\pi$

**c.** Amplitude 2, period $\frac{3}{2}\pi$

**d.** Amplitude 2, period $2\pi$

Amplitude $= |A| = 2$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

**45** [Challenging] • Target: Graphing with transformations • Cross-Lesson: 1.02 Domain and Range

What is the amplitude and period of $y = 2\sin(3x) + 2$?

**a.** Amplitude 3, period $\frac{2}{3}\pi$

**b.** Amplitude 2, period $2\pi$

**c. Amplitude 2, period $\frac{2}{3}\pi$ ✓**

**d.** Amplitude 2, period $\frac{3}{2}\pi$

Amplitude $= |A| = 2$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

© ClipSAT — clipsat.org | All Rights Reserved